

Project Design Phase

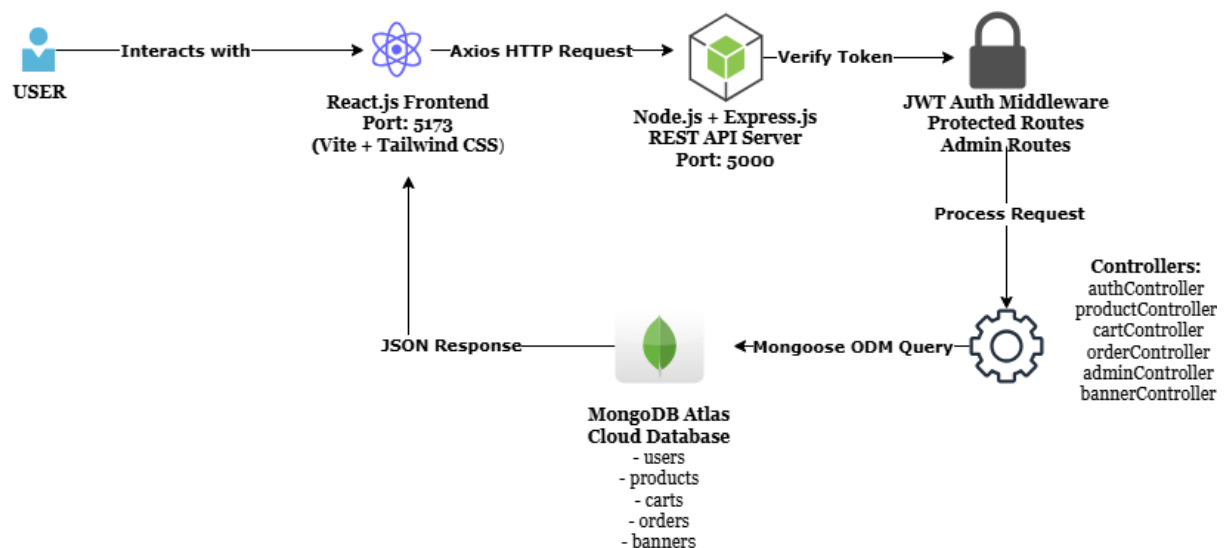
Solution Architecture

Date	15 February 2025
Prepared By	Sowmya Kella
Project Name	ShopEZ: E-Commerce Application
Maximum Marks	4 Marks

Solution Architecture:

The Solution Architecture Diagram of ShopEZ illustrates the complete data flow between all layers of the application. The user interacts with the React.js frontend which sends HTTP requests via Axios to the Node.js and Express.js REST API backend. All requests pass through JWT Authentication Middleware before reaching the controllers which process the business logic and communicate with MongoDB Atlas cloud database using Mongoose ODM. The database returns the data as a JSON response back to the frontend for display.

Solution Architecture Diagram:



The ShopEZ solution architecture follows a three tier MERN stack structure combined with the MVC (Model View Controller) design pattern that ensures clean separation of concerns between the frontend, backend and database layers. The React.js frontend built with Vite and Tailwind CSS runs on port 5173 and acts as the View layer, handling all user interactions and displaying data to the user. It communicates with the backend through Axios HTTP requests. The Node.js and Express.js REST API server runs on port 5000 and acts as the Controller layer that processes all incoming requests and applies business logic. All protected routes pass through the JWT Authentication Middleware which verifies the user token before allowing access to any secured endpoint. Once verified, the request is passed to the appropriate

controller — authController, productController, cartController, orderController, adminController or bannerController — depending on the type of request.

The controllers interact with the MongoDB Atlas cloud database through Mongoose ODM to perform create, read, update and delete operations on the five collections — users, products, carts, orders and banners. These collections act as the Model layer — defining the structure and relationships of all data in the application. The database returns the requested data which is sent back to the frontend as a JSON response for display to the user. This MVC based MERN architecture ensures that ShopEZ is secure, scalable and maintainable — with each layer having a clear and dedicated responsibility in the overall system.